CLIENT–SERVER ARCHITECTURE

Task 2 — Frontend, Backend, and Database Communication

Architecture Diagram

HOW COMMUNICATION WORKS

1. The frontend sends a request to the backend.
2. The backend processes it and accesses the database when needed.
3. The backend returns a response to the frontend.

The frontend normally does not connect directly to the private database.

Explanation

Frontend / Client: The user-facing part of the application. It displays pages and sends requests when users interact with the website.

Backend / Server: The application logic, built here with Python and Django. It receives requests, processes them, and communicates with the database.

Database: Stores application information. The backend sends queries to read or update data and receives the results.

Example: When a student clicks “View Result,” the frontend sends a request to the backend. The backend retrieves the result from the database and sends it back so the browser can display it.